

Multi-Recognizer Multi-Attacker Adversarial Training for Robust Face Recognition

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Repository: <https://github.com/ZeroOreos/FYP>

Abstract—Face recognition systems are used for identity verification and identification, and are assessed through a suite of protocols ranging from clean benchmarks to robustness-oriented evaluations. Practical use of face recognition models requires real-world robustness considerations, where appearance variation, dirty-data conditions, and adversarial inputs require identity embeddings to remain both separable and stable. Recent adversarial ensemble training efforts have attempted to address part of this problem by increasing attack-side pressure during training. Following such efforts, this work investigates whether adding recognizer diversity improves robustness beyond attack diversity alone. To test this, a single ArcFace/iResNet100 target is trained on WebFace4M under five comparison conditions: clean training, recognizer-only pressure, single-attack training, attack-ensemble training, and full recognizer-plus-attacker training. The central comparison holds the attacker suite fixed to PGD, BPFA, and DEANet, then adds auxiliary ArcFace, CosFace, and CurricularFace recognizer heads during training only. This raises average robust accuracy from 52.32% to 56.55% and reduces average attack success from 44.32% to 39.45%, while verification accuracy decreases from 98.33% to 98.21% and EER increases from 1.78% to 1.93%. Inference cost is unchanged because only the target recognizer is deployed, but training cost increases substantially. The result is a calibrated tradeoff: recognizer diversity adds measured digital robustness beyond attack diversity alone, but moves the model along a robustness–verification–training-cost frontier.

Index Terms—face recognition, adversarial training, ensemble robustness, ArcFace, verification, biometric security

I. INTRODUCTION

Face recognition (FR) systems are used for identity verification and identification, and are commonly assessed through clean benchmarks such as LFW, MegaFace, and IJB-C [11]–[13]. Deep models and margin-based objectives, including DeepFace, FaceNet, VGGFace, SphereFace, and ArcFace, have made these protocols strong references for clean recognition performance [1]–[5]. Clean accuracy, however, does not fully describe deployment reliability. FR systems may encounter distribution shift, appearance variation, common corruptions, and deliberate adversarial inputs. This has led to dirty-data and robustness-oriented benchmarks, including corrupted or appearance-varied face evaluations and fine-grained adversarial test frameworks [23], [26]. Decision-based black-box attacks have also been demonstrated against FR systems [21], and surveys document diverse attack and defense settings [22]. For biometric systems, these failures matter because the

same embedding must support both high identity separability and stable decisions under non-clean inputs.

Adversarial training is one way to improve reliability under such pressure, but robustness studies repeatedly show that results depend on the training and evaluation threat model. Robust optimization frames defense as a min–max problem, ensemble adversarial training reduces dependence on a single adversary, and evaluation suites such as AutoAttack, RobustBench, and MultiRobustBench emphasize that narrow attack coverage can overestimate robustness [18]–[20], [24], [25]. In FR, this concern is tied to verification: the model is not only predicting a training identity, but learning an embedding space whose pairwise distances are thresholded at test time. A training signal that improves adversarial classification can still alter the clean verification geometry.

This suggests two sources of training pressure. Attack diversity determines which perturbation procedures the target is exposed to. Recognizer-objective diversity determines which embedding geometries judge the target during training. Modern margin and quality-aware objectives show that changing the recognition loss changes the difficulty structure of the embedding space [5]–[10]. The question is therefore not only whether a stronger attacker helps, but whether recognizer-side diversity adds robustness once attack diversity is already present. The same added pressure may also reduce verification quality or increase training cost.

The practical setting matters because large-scale FR is already resource intensive. RetinaFace-style alignment, WebFace-scale data, Partial FC, and sub-center ArcFace address alignment, many identities, classifier-head memory, and noisy web labels [14]–[17]. Adding online attacks and auxiliary recognizer objectives compounds these costs. For that reason, this paper treats robustness as a tradeoff frontier rather than a free improvement: robust accuracy, verification accuracy, equal error rate, memory pressure, batch size, and wall-clock cost must be read together.

This work studies whether recognizer-objective diversity adds measurable value after attacker diversity is already present. The experiments use one deployed target recognizer and compare clean training, recognizer-only pressure, single-attack training, attack-ensemble training, and the full recognizer-plus-attacker setting. The goal is not to claim state-of-the-art FR, but to isolate a specific design effect: whether training-time recognizer pressure can improve digital adver-

serial robustness beyond a multi-attack baseline, and what that improvement costs in verification quality and training efficiency.

II. METHODS

A. Design Rationale

The design separates the deployed recognizer from the training pressures used to shape it. The deployed model is a single ArcFace/iResNet100 target. During training, the target sees adversarial samples from several attack families and auxiliary recognition objectives that evaluate the same clean or perturbed faces. This keeps inference cost unchanged while testing whether the target can absorb useful pressure from a broader training environment.

The sets are intentionally compact. ArcFace supplies the deployed angular margin target, CosFace supplies a cosine-margin alternative, and CurricularFace changes the weighting of hard examples. PGD supplies direct optimization, BPFA supplies transfer-oriented pressure, and DFANet supplies FR feature-level pressure. The full condition is therefore one recognizer trained under rotating attacker pressure and training-only recognizer pressure, not a deployed multi-model ensemble.

B. Training Objective

The defended target is an ArcFace model with an iResNet100 backbone and a 512-dimensional embedding. Training uses WebFace4M manifests with RetinaFace-style aligned 112×112 crops. The configured subset contains 205,849 identities, selected with a fixed random seed and a minimum of two images per identity, so all comparison conditions share the same identity universe.

Let $f_\theta(x) \in \mathbb{R}^{512}$ denote the target embedding and $g_\theta(x) = f_\theta(x)/\|f_\theta(x)\|_2$ its normalized form. A margin head with normalized class weights W produces logits of the form

$$\ell_y(x) = s \phi_y(W_y^\top g_\theta(x)), \quad \ell_j(x) = s W_j^\top g_\theta(x), \quad j \neq y, \quad (1)$$

where $\phi_y(\cdot)$ is the margin rule applied to the ground-truth class. The auxiliary recognizers differ mainly through ϕ_y , giving the same samples different margin geometries rather than redundant copies of the target classifier.

Adversarial examples are generated by a family-conditioned inner problem

$$x_k^{adv} \approx \arg \max_{\|\delta\|_\infty \leq \epsilon_k} \mathcal{J}_k(x + \delta, y; \theta, \mathcal{S}), \quad (2)$$

where $k \in \{\text{PGD}, \text{BPFA}, \text{DFANet}\}$ and $\mathcal{S}$ is the available surrogate set. The optimized loss combines the pressure terms separated by the ablation:

$$\mathcal{L} = \lambda_c \mathcal{L}_c + \lambda_a \mathcal{L}_a + \lambda_r \mathcal{L}_r + \lambda_s \mathcal{L}_{sim} + \lambda_h \mathcal{L}_{hard}. \quad (3)$$

Here $\mathcal{L}_c = \text{CE}(\ell(x), y)$ and $\mathcal{L}_a = \text{CE}(\ell(x^{adv}), y)$ keep the target discriminative, while

$$\mathcal{L}_r = \sum_{r \in \mathcal{R}} \alpha_r [\text{CE}(\ell_r(x), y) + \text{CE}(\ell_r(x^{adv}), y)] \quad (4)$$

adds pressure from recognizer heads $\mathcal{R} = \{\text{ArcFace}, \text{CosFace}, \text{CurricularFace}\}$. The consistency term discourages excessive clean/adversarial embedding drift,

$$\mathcal{L}_{sim} = 1 - \cos(g_\theta(x), g_\theta(x^{adv})), \quad (5)$$

and $\mathcal{L}_{hard}$ emphasizes mined hard pairs or hard samples. The recognizer ensemble affects optimization only. After training, only f_θ is deployed.

C. Training Pipeline

Training follows a three-stage curriculum over 24 epochs. Clean warmup lasts 3 epochs, shallow adversarial pressure lasts 9 epochs, and full adversarial pressure lasts 12 epochs. Shallow and full stages use the same attack families with different strengths. For stage t , each batch is split as

$$B_t = B_t^{clean} \cup B_t^{adv}, \quad |B_t^{adv}| = \rho_t |B_t|, \quad (6)$$

where $\rho_t = 0$ during clean warmup, increases during shallow training, and is 0.50 in the full condition. One primary attacker family is selected in round-robin order for each adversarial batch. Impersonation and dodging intents are balanced to cover the main FR failure modes.

Each iteration has a target branch and pressure branches. Clean images update the ArcFace/iResNet100 target through the margin objective. During adversarial stages, part of the batch enters the attack branch, where the selected attacker is optimized against the current target or surrogate state with configured random starts and restart selection. The resulting adversarial images return to the target branch for adversarial classification and clean/adversarial embedding consistency.

Recognizer pressure is attached after the target representation is formed. ArcFace, CosFace, and CurricularFace heads evaluate the same clean or adversarial samples and contribute weighted losses. The target Partial FC rate is 0.30, while auxiliary heads use 0.12 to limit classifier-head memory. Attack generation is chunked to fit the iResNet100 target and Partial FC head on 24 GB GPUs. The hard-sample term is staged so early epochs learn identity structure before mined pairs receive stronger weight.

D. Comparison Sequence

The comparison sequence isolates one source of pressure at a time. *Baseline-clean* sets the clean recognition reference. *Recognizer-ensemble-only* tests whether auxiliary recognizer pressure changes embedding quality without attacks. *Single-attack-only* tests the risk of narrow adversarial training. *Attack-ensemble-only* measures the multiattack baseline. *Full-ensemble* then adds recognizer diversity on top of the same attacker suite, making it the decisive comparison for the main claim.

E. Threat Model and Evaluation Boundary

The evaluated attacks are digital image-space attacks, not physical or presentation attacks. PGD is treated as white-box optimization, BPFA as transfer-oriented pressure, and DFANet as an FR feature-level attack. During training, attacks are

generated against the current target/surrogate state, so the defense is not based on hiding the target. Robust classification is identification-style: it measures whether the closed-set identity prediction survives perturbation. Pair verification measures a different property: whether embeddings still separate same-identity from different-identity pairs.

The evaluation is not adaptive. It does not include an adversary that optimizes through the complete training-time recognizer ensemble, tunes attacks after observing all defense hyperparameters, or performs query-based score/decision attacks against the deployed system. The claim is therefore restricted to the PGD/BPFA/DFANet digital suite and to the ablation question of whether recognizer-objective pressure adds robustness once attacker diversity is already present.

F. Experimental Setup and Metrics

All main experiments use ArcFace/iResNet100 on WebFace4M with 4,015,275 training images and 219,826 validation images over 205,849 identities. Partial FC uses a 0.30 negative sampling rate to keep large-class training feasible on 8 RTX 4090D GPUs. Main per-GPU batch settings are clean warmup 128, shallow adversarial 48, full adversarial 32, and attack chunk size 32. Mixed precision, channels-last tensors, distributed training, and manifest-backed dataloading are used.

Metrics are grouped by the property they measure. Closed-set validation reports identity classification. Robust validation reports adversarial accuracy, ASR, clean-to-robust drop, and per-attack behavior. For attack family k ,

$$A_k^{rob} = \frac{1}{N} \sum_i \mathbf{1}[\hat{y}(x_{i,k}^{adv}) = y_i], \quad \text{ASR}_k = 1 - A_k^{rob}. \quad (7)$$

Verification uses cosine scores $s_{ij} = g_\theta(x_i)^\top g_\theta(x_j)$ and reports pair accuracy, EER, ROC-AUC, TAR at low FAR, and cosine separation. A pair is accepted at threshold τ when

$$d_\tau(x_i, x_j) = \mathbf{1}[s_{ij} \geq \tau]. \quad (8)$$

For genuine pairs $\mathcal{P}^+$ and impostor pairs $\mathcal{P}^-$,

$$\text{TAR}(\tau) = \frac{1}{|\mathcal{P}^+|} \sum_{(i,j) \in \mathcal{P}^+} d_\tau(x_i, x_j), \quad (9)$$

$$\text{FAR}(\tau) = \frac{1}{|\mathcal{P}^-|} \sum_{(i,j) \in \mathcal{P}^-} d_\tau(x_i, x_j). \quad (10)$$

EER is the operating point where false accept and false reject rates are equal. The verification protocol is in-domain because pairs are sampled from the WebFace4M training manifest and validation identities overlap training identities. These numbers therefore analyze embedding geometry rather than external generalization. Pair images are deduplicated, embeddings are extracted in distributed GPU batches, cosine scores are chunked, and ROC statistics are computed from sorted scores. Training cost is reported with wall time, GPU-hours, throughput, attack-generation share, peak CUDA memory, and active per-GPU batch sizes. Inference cost is unchanged because auxiliary recognizers and attackers are discarded after training.

TABLE I

CLOSED-SET CLEAN AND ADVERSARIAL ROBUSTNESS RESULTS. ROBUST AVERAGE IS OVER PGD/BPFA/DFANet UNLESS MARKED OTHERWISE.

Condition	Clean Acc.	Robust Acc.	Drop	ASR
Baseline-clean	99.23	0.21	99.02	99.79
Recognizer-only	99.23	0.23	99.01	99.77
Single-attack	79.69	34.51	45.18	55.09
Attack-ensemble	93.98	52.32	41.65	44.32
Full-ensemble	93.40	56.55	36.85	39.45

All numbers are percentages. Robust accuracy and ASR are evaluated over PGD/BPFA/DFANet.

Several implementation choices make the full setting computationally feasible. Recognizer pressure is stage-gated so clean warmup remains fast. Auxiliary recognizer heads use a lower Partial FC sampling rate than the target, attack batches are chunked, verification is batched and distributed, and the full-adversarial batch size is kept at 32 after larger batch trials increased memory pressure without improving stable throughput. Training attackers and evaluation attackers are separated in configuration. Clean-only and recognizer-only conditions receive no adversarial training pressure, but the final robustness table evaluates all conditions against the same PGD/BPFA/DFANet suite.

The evaluation suite remains narrower than the broader deployment threat model. Stronger AutoAttack/APGD-style checks [20], TransMix-style transfer attacks [30], certified FR robustness tests [31], AdvFaceGAN, Adv-Makeup, Greedy-DiM morphing, diffusion-based attacks, and physical or PAD-aware stress tests are not counted as evidence here because they require faithful separate implementations or generated-image protocols [32]–[36].

III. RESULTS

A. Overview

The results support a conditional claim rather than a simple ranking. Recognizer pressure alone preserves clean in-domain verification but does not make the target robust. Narrow PGD-only pressure improves some adversarial behavior but damages both classification and verification. The full method becomes useful only after the attacker side is already diversified: recognizer-objective pressure improves robustness beyond the attack ensemble, while moving the embedding away from the clean-verification optimum.

B. Robustness

Table I holds the attacker suite fixed for the key comparison. Adding recognizer-objective pressure on top of PGD, BPFA, and DFANet raises robust accuracy from 52.32% to 56.55%, lowers ASR from 44.32% to 39.45%, and reduces the clean-to-robust drop from 41.65 to 36.85 points. The remaining difference is therefore attributable to recognizer diversity rather than attack coverage.

The control rows clarify the boundary. Clean and recognizer-only models remain almost fully vulnerable under

TABLE II
IN-DOMAIN WEBFACE4M PAIR-VERIFICATION METRICS. TAR IS
REPORTED AT LOW FAR OPERATING POINTS, FOLLOWING STANDARD FR
REPORTING PRACTICE.

Condition	Verif. Acc.	EER	ROC-AUC	TAR@1e-4	TAR@1e-5
Baseline-clean	99.96	0.0470	99.999	99.86	99.65
Recognizer-only	99.96	0.0410	99.999	99.92	99.73
Single-attack	89.74	12.82	92.01	0.37	0.11
Attack-ensemble	98.33	1.78	99.75	85.02	73.38
Full-ensemble	98.21	1.93	99.73	84.81	74.16

All numbers are percentages. Lower EER is better; higher values are better for all other columns.

the three-attack suite, with robust accuracy near 0.2000% and ASR near 99.80%. Recognizer pressure alone therefore does not create adversarial robustness. Single-attack training is also incomplete. Its three-attack average is 34.51%, with BPFA at 45.71% but PGD and DFANet both near 28.90%, showing weak transfer across attack families.

The per-attack structure is more informative than the average alone. PGD rises from 45.45% to 52.55%, BPFA from 66.09% to 69.56%, and DFANet from 45.43% to 47.55%. The largest gain is on PGD, suggesting that recognizer pressure most strongly affects direct margin optimization. DFANet improves least, consistent with feature-level FR attacks already operating close to the embedding objective being regularized.

The clean-accuracy movement shows the cost of that gain. Full-ensemble clean accuracy is slightly lower than attack-ensemble-only (93.40% versus 93.98%), so the recognizer heads do not act as simple clean regularizers. They appear to move the representation toward perturbation stability. The single-attack row is more severe: PGD-only training reaches 79.69% clean accuracy and 34.51% average robust accuracy, consistent with over-specialized adversarial pressure degrading the margin classifier while leaving substantial cross-attack vulnerability.

The epoch histories suggest that this clean-accuracy cost is schedule sensitive, but not simply unresolved undertraining. In the full condition, clean validation recovers from 87.57% at epoch 16 to 92.37% at epoch 20 and 93.40% at epoch 24, while robust accuracy rises from 46.15% to 55.34% and then 56.55%. The late gains therefore taper after the learning-rate drop. Longer training or a lower-pressure finetuning stage might recover additional clean accuracy, but the observed trend does not show that the full gap to attack-ensemble-only would disappear without changing the schedule.

C. Verification

Table II measures embedding geometry rather than external generalization. The verification bundle contains 409,196 WebFace4M pairs from the training manifest, and validation identities overlap training identities. Near-ceiling clean verification is therefore expected. The informative signal is the relative degradation across conditions, especially at low-FAR operating points and in cosine separation.

The clean and recognizer-only rows provide the control. Both retain about 99.96% verification accuracy, near-perfect

ROC-AUC, and EER below 0.0500%. The recognizer-only condition slightly improves EER and low-FAR TAR, indicating that auxiliary margin heads are not inherently harmful to the embedding space. Degradation appears when margin diversity is coupled to adversarial image pressure, where the target must satisfy clean identity separation and perturbed-feature stability at the same time.

The single-attack row shows the strongest geometric failure: 89.74% verification accuracy, 12.82% EER, 92.01% ROC-AUC, and only 0.37% TAR@1e-4. This low-FAR collapse indicates a score-distribution shift, not only a loss of class accuracy. For FR, closed-set robust accuracy is therefore insufficient without verification metrics.

Attack-ensemble training restores much of this structure while providing robustness: 98.33% verification accuracy, 1.78% EER, and 99.75% ROC-AUC. The full ensemble then improves robust accuracy at a small but consistent verification cost: 98.21% accuracy, 1.93% EER, and TAR@1e-4 of 84.81%. The mean negative cosine increases from 0.0440 to 0.0640, indicating weaker negative-pair separation. The full model is harder to attack, but part of the clean verification geometry is compressed by the added invariance pressure.

The verification trend gives a similar schedule signal. Full-ensemble verification partially recovers late in training, moving from 97.70% accuracy and 2.43% EER at epoch 21 to the final 98.21% accuracy and 1.93% EER. Attack-ensemble-only also improves late, reaching 98.33% accuracy and 1.78% EER. This makes the verification loss better interpreted as a schedule-sensitive tradeoff than as a failed convergence point. A longer low-learning-rate tail, delayed recognizer pressure, or annealed recognizer weights could plausibly improve pair geometry, but each option changes the robustness-cost balance that the current ablation measures.

The full method therefore moves the model along a robustness-verification frontier. Where adversarial probes are plausible, the 4.23-point robust accuracy gain and 4.87-point ASR reduction may justify the roughly 0.12-point verification-accuracy loss and 0.15-point EER increase relative to attack-ensemble-only. Where clean verification dominates the operating requirement, the clean or recognizer-only rows remain preferable. This is the main interpretive result: multi-recognizer multi-attacker training should be viewed as targeted robustness pressure rather than a replacement for clean FR training.

D. Training Cost

Table III reports the price of stronger pressure under a memory-capped batch schedule. The clean baseline takes 12.70 wall-clock hours. Recognizer-only training reaches 26.10 hours because auxiliary recognizer heads add backbone and classifier-head passes even without attacks. Single-attack training takes 37.60 hours, attack-ensemble training takes 75.50 hours, and the full condition reaches 110.1 hours, or 880.7 GPU-hours. Wall time combines algorithmic cost with batch-size constraint. Clean and recognizer-only conditions train at batch 128, while adversarial phases are capped at 48

TABLE III

TRAINING COST MEASURED FROM COMPLETED 24-EPOCH HISTORIES ON 8 GPU'S. WALL TIME INCLUDES TRAINING, VALIDATION, VERIFICATION, AND MODEL-SAVING OVERHEAD. BATCH LISTS THE DOMINANT ACTIVE PER-GPU BATCH SIZE.

Condition	Batch	Wall h	GPU h	Img/s	Attack Share	Peak GB
Baseline-clean	128	12.70	101.9	262.8	0.0000	20.30
Recognizer-only	128	26.10	208.9	128.1	0.0000	20.60
Single-attack	48/32	37.60	300.9	89.00	55.70	20.30
Attack-ensemble	48/32	75.50	604.2	44.30	74.90	20.30
Full-ensemble	48/32	110.1	880.7	32.70	67.70	22.60

Img/s is end-to-end average throughput over the completed training history. Attack share is attack generation as a fraction of measured train compute.

and 32. The clean baseline therefore uses up to four times the full-adversarial batch size for a substantial part of training.

The dominant cost is the adversarial inner loop. Attack-ensemble-only spends 74.90% of measured train compute on attack generation, and the full ensemble still spends 67.70% there despite the added recognizer losses. Each attacked batch performs repeated target/surrogate forward-backward passes before the ordinary parameter update; the full condition also evaluates recognizer heads and samples large-class margin weights. Peak memory reaches 22.60 GB on 24 GB cards, so the method is limited by both repeated computation and reduced batch size. The measured bottleneck is repeated adversarial optimization under a large-class FR head.

The full ensemble gains 4.23 robust-accuracy points over attack-ensemble-only at the cost of 34.60 additional wall-clock hours and 276.50 GPU-hours. The marginal robustness gain is therefore expensive. Existing acceleration work suggests practical directions: recycling gradients or adversarial states as in Free Adversarial Training [27], reducing repeated full backpropagation as in YOPO [28], or using cheaper single-step adversaries with safeguards against catastrophic overfitting [29]. For this FR setting, the closest direct ablations are lower-frequency recognizer pressure, cached or replayed adversarial seeds, cheaper early-epoch attack curricula, shared surrogate forwards across attacks, and selective full-depth attack refreshes. These target the measured bottleneck without changing the multi-recognizer multi-attacker objective.

Schedule changes are another path, but they trade runtime for cleaner optimization dynamics. Extending the final low-learning-rate phase could test whether verification continues to recover after robustness has mostly stabilized. Delaying recognizer pressure until the target has stronger adversarial features could reduce early embedding distortion. Annealing the recognizer loss late in training could preserve some robustness while allowing the target margin head to re-separate clean pairs. These variants are scientifically useful, but the full run already costs 880.7 GPU-hours, so each additional schedule test should be treated as a deliberate robustness, verification, and compute tradeoff rather than a free correction.

IV. DISCUSSION AND CONCLUSION

The work is best read as an ablation-backed framework result. It does not show state-of-the-art FR, prove that every

ensemble member is beneficial, or establish deployment robustness against all attacks. It shows a narrower but defensible effect: when a target already receives multiattack pressure, adding recognizer-objective diversity improves measured robustness under the same attacker suite. The single-attack result also supports the need for broader attack-family training rather than narrow PGD-only pressure.

Several limitations define the boundary of this result. The attacker suite is explicit but not exhaustive. Verification and closed-set robust classification answer different questions, so both are required for FR robustness analysis. The verification protocol is in-domain; external unseen-identity benchmarks such as LFW, CFP-FP, AgeDB, or IJB-style evaluations are needed before deployment claims. The recognizer ensemble is training-time pressure, not a deployment ensemble. Training cost also shapes the design itself: memory-limited batch sizes, repeated attack forwards, and large-class margin heads make the full condition much slower than the clean baseline. Finally, the work does not yet establish robustness to adaptive ensemble attacks, modern semantic or generative attacks, morphing attacks, corruptions, OOD appearance variation, or physical/PAD-aware attacks.

A. Conclusion

This study implements a face-recognition robustness framework that trains one ArcFace/iResNet100 target under explicit recognizer and attacker ensemble pressure. The result is an ablation-backed tradeoff rather than a leaderboard claim. Recognizer-objective diversity adds measured adversarial robustness beyond attacker diversity, but the gain comes with lower clean-verification performance and a large training-cost increase. The method is therefore best understood as a practical training framework for FR hardening: useful when digital adversarial robustness is part of the operating requirement, less attractive when clean verification is the only priority, and still bounded by the attacks and in-domain verification protocol evaluated here.

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